

A ZFC positive answer for overcomplete sets in ordinal interval spaces below ω_2

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Status. Candidate full solution, likely valid subject to expert review. This packet proves a ZFC positive answer to Question 48 from Koszmider's arXiv:2006.00806: for every ordinal $\xi < \omega_2$, the Banach space $C([0, \xi])$ admits an overcomplete set.

1 Source question

The source paper is:

P. Koszmider, *On the existence of overcomplete sets in some classical nonseparable Banach spaces*, arXiv:2006.00806.

On page 25, Question 48 asks:

Can one prove in ZFC that the Banach spaces $C([0, \xi])$ for all ordinals $\xi < \omega_2$ admit overcomplete sets?

There are also some natural open questions here which concern more particular classes for Banach spaces like the following three questions:

Question 48. *Can one prove in ZFC that the Banach spaces $C([0, \xi])$ for all ordinals $\xi < \omega_2$ admit overcomplete sets?*

Partial results related to Question 48 are Theorem 11 (2) and result of 37 (Theorem 5 (3) in this paper). They cover all remaining ordinals as $C([0, \xi])$ admits a fundamental biorthogonal system $(1_{[0, \eta]}, \delta_\eta - \delta_{\eta+1})_{\eta < \xi}$ and $1_{[0, \xi]}, \delta_\xi$.

2 Definitions and criterion

Definition 1. *Let X be a Banach space. A set $D \subseteq X$ is overcomplete if $|D| = \text{dens}(X)$ and every subset $A \subseteq D$ with $|A| = |D|$ has dense linear span in X .*

We shall use the following elementary criterion repeatedly.

Lemma 2. *Let X be a Banach space of density ω_1 . If there is a family $(x_\alpha)_{\alpha < \omega_1} \subseteq X$ such that*

$$\{\alpha < \omega_1 : x^*(x_\alpha) = 0\}$$

is countable for every nonzero $x^ \in X^*$, then $\{x_\alpha : \alpha < \omega_1\}$ is overcomplete.*

Proof. The family has cardinality ω_1 after deleting possible repetitions; in the constructions below the points are explicitly distinct. If an uncountable subfamily had non-dense linear span, Hahn–Banach would give a nonzero functional vanishing on it, contradicting the hypothesis. $\square$

We also use the almost coherent injections from Koszmider’s proof. There are injections $e_\alpha : \alpha \rightarrow \mathbb{N}$ for $\alpha < \omega_1$, with $\mathbb{N} = \{0, 1, 2, \dots\}$, such that

$$e_\beta \upharpoonright \alpha =^* e_\alpha \quad (\alpha < \beta < \omega_1),$$

and with $\mathbb{N} \setminus e_\alpha[\alpha]$ infinite. Consequently, if $A \subseteq \omega_1$ is uncountable and $\gamma < \omega_1$, then there are an uncountable $A' \subseteq A \setminus (\gamma + 1)$ and an injection $g : \gamma + 1 \rightarrow \mathbb{N}$ such that

$$e_\alpha \upharpoonright (\gamma + 1) = g \quad (\alpha \in A').$$

3 Proof Intuition

Koszmider’s original construction handles $C([0, \omega_1])$ by turning the values of a fixed functional on an uncountable subfamily into zeros of one analytic function. The earlier lane-16 packets pushed this through finite ω_1 -block sums and then through the first countable block $C([0, \omega_1 \cdot \omega])$ by assigning disjoint Taylor exponent sets to the countably many active blocks.

The remaining difficulty is the ω_1^2 -level. The interval $[0, \omega_1^2]$ has a closed skeleton

$$B = \{\omega_1 \eta : \eta \leq \omega_1\}$$

homeomorphic to $[0, \omega_1]$, and gaps between consecutive skeleton points which are copies of the open part of $[0, \omega_1]$. We split a function into a skeleton part plus a gap part vanishing on the skeleton. The skeleton part uses the strengthened one-block analytic construction. The gap part uses endpoint-vanishing block functions, so no endpoint mass has to be detected inside a gap; endpoint masses belong to the skeleton functional. For each selected vector, only countably many gaps are perturbed, and the coherent injection makes the perturbation norms tend to zero near every limit skeleton point. Thus the vector is continuous on $[0, \omega_1^2]$.

For any fixed functional, only countably many gaps and countably many points in those gaps matter. One stabilization of the coherent injections freezes all of them. The resulting evaluation is an analytic function whose Taylor coefficients live in pairwise disjoint exponent sets: one set for the skeleton and one set for each active gap. Coefficient comparison forces the skeleton functional and every active gap functional to vanish. Hence every nonzero functional has only countably many zeros on the constructed family.

4 The ω_1^2 representative

Let $L = [0, \omega_1^2]$. For $\eta \leq \omega_1$, write

$$s_\eta = \omega_1 \eta.$$

The skeleton $B = \{s_\eta : \eta \leq \omega_1\}$ is homeomorphic to $[0, \omega_1]$.

There is a norm-one linear extension $E : C(B) \rightarrow C(L)$ defined as follows:

$$(Ey)(s_\eta) = y(s_\eta) \quad (\eta \leq \omega_1),$$

and, for $s_\eta < t \leq s_{\eta+1}$,

$$(Ey)(t) = y(s_{\eta+1}).$$

This is continuous because if $\eta_i + 1 \rightarrow \lambda$ in the skeleton, then $y(s_{\eta_i+1}) \rightarrow y(s_\lambda)$; at the right endpoint of each gap the extension is eventually constant. At a skeleton point s_λ , the immediate successor $s_\lambda + 1$ gives right-isolating neighbourhoods, so only the left-hand approach through earlier gaps has to be checked. Let

$$W = \{f \in C(L) : f|_B = 0\}.$$

Then $C(L) = E[C(B)] \oplus W$.

Identify $C(B)$ with $C([0, \omega_1])$. Put

$$u_\alpha = \mathbf{1}_{[0, \alpha]} \in C([0, \omega_1]) \quad (\alpha < \omega_1),$$

and

$$Z = \{z \in C([0, \omega_1]) : z(0) = z(\omega_1) = 0\}.$$

For $\eta < \omega_1$, let $\iota_\eta : Z \rightarrow W$ be the insertion into the closed gap $[s_\eta, s_{\eta+1}]$, using the order homeomorphism with $[0, \omega_1]$, and zero outside that gap.

Theorem 3. *In ZFC, $C([0, \omega_1^2])$ admits an overcomplete set.*

Proof. Choose pairwise disjoint infinite sets

$$P_\infty, P_0, P_1, \dots \subseteq \mathbb{N}$$

and enumerations $p_j : \mathbb{N} \rightarrow P_j$, $j \in \{\infty, 0, 1, \dots\}$, such that

$$p_j(k) \geq 2k + 1 \quad (k < \omega).$$

Set

$$q_\infty(t) = \sum_{k=0}^{\infty} 2^k t^{p_\infty(k)} \quad (|t| < 1/\sqrt{2}).$$

Fix $0 < b < 1/2$, choose distinct $r_\alpha \in (b/2, b)$ for $\alpha < \omega_1$, and choose positive scalars $a_m \downarrow 0$.

For $\alpha < \omega_1$, define the skeleton vector $y_\alpha \in C(B) \cong C([0, \omega_1])$ by

$$y_\alpha = r_\alpha^{p_\infty(0)} u_\alpha + \sum_{\beta < \alpha} r_\alpha^{p_\infty(e_\alpha(\beta)+1)} (u_\beta - u_\alpha) + q_\infty(r_\alpha) \mathbf{1}.$$

For $m < \omega$, define the endpoint-vanishing gap vector

$$z_\alpha^{(m)} = \sum_{\beta < \alpha} r_\alpha^{p_m(e_\alpha(\beta)+1)} (u_\beta - u_\alpha) \in Z.$$

The series converge absolutely because e_α is injective and $\sum_k b^{p_j(k)} < \infty$.

Now put

$$x_\alpha = Ey_\alpha + \sum_{\eta < \alpha} a_{e_\alpha(\eta)} \iota_\eta(z_\alpha^{(e_\alpha(\eta))}) \quad (\alpha < \omega_1).$$

The displayed sum is well-defined pointwise because each point of $L \setminus B$ lies in one gap. It is continuous: at successor skeleton points the inserted gap functions vanish at both endpoints; at a limit skeleton point s_λ , the finite sets $\{\eta < \alpha : e_\alpha(\eta) \leq M\}$ are bounded below λ , so $a_{e_\alpha(\eta)} \rightarrow 0$ as $\eta \rightarrow \lambda$. The same argument handles the top point s_{ω_1} , while the right side of each skeleton point is isolated by its immediate successor. Thus $x_\alpha \in C(L)$. Also the x_α 's are distinct, since

$$x_\alpha(s_{\omega_1}) = q_\infty(r_\alpha),$$

and q_∞ is strictly increasing on $(0, b)$.

We prove the functional separation property. Let $\mu \in C(L)^*$ and suppose that

$$\mu(x_\alpha) = 0 \quad (\alpha \in A)$$

for some uncountable $A \subseteq \omega_1$. Since L is scattered, μ is an atomic measure with countable support.

Let $\nu = E^* \mu \in C(B)^*$. Let $S \subseteq \omega_1$ be the countable set of gaps on which the restriction of μ to the open gap $(s_\eta, s_{\eta+1})$ is nonzero. For $\eta \in S$, view this restriction as a measure μ_η on $[0, \omega_1]$ with no mass at 0 or ω_1 ; it represents the induced functional on Z .

Choose $\gamma < \omega_1$ large enough so that:

- $S \subseteq [0, \gamma]$;
- the support of ν below the top point of B is contained in $[0, \gamma]$;
- for every $\eta \in S$, the support of μ_η in $[0, \omega_1]$ is contained in $[0, \gamma]$.

Stabilize the coherent injections on $\gamma + 1$: pass to an uncountable $A' \subseteq A \setminus (\gamma + 1)$ and an injection $g : \gamma + 1 \rightarrow \mathbb{N}$ such that $e_\alpha \upharpoonright (\gamma + 1) = g$ for $\alpha \in A'$.

For the skeleton functional ν , write

$$b_\beta = \nu(u_\beta) \quad (\beta \leq \gamma), \quad d = \nu(\mathbf{1}),$$

and let c be the eventual value of $\nu(u_\xi)$ for $\xi > \gamma$. For each $\eta \in S$, write

$$b_{\eta, \beta} = \mu_\eta(u_\beta) \quad (\beta \leq \gamma),$$

and let c_η be the eventual value of $\mu_\eta(u_\xi)$ for $\xi > \gamma$.

For $\alpha \in A'$, with $t = r_\alpha$, the equality $\mu(x_\alpha) = 0$ becomes $F(t) = 0$, where

$$\begin{aligned} F(t) = & c t^{p_\infty(0)} + \sum_{\beta \leq \gamma} (b_\beta - c) t^{p_\infty(g(\beta)+1)} + d q_\infty(t) \\ & + \sum_{\eta \in S} a_{g(\eta)} \sum_{\beta \leq \gamma} (b_{\eta, \beta} - c_\eta) t^{p_{g(\eta)}(g(\beta)+1)}. \end{aligned}$$

This is analytic on $|t| < 1/\sqrt{2}$. The last sum converges normally on compact subdiscs because $\sum_{\eta \in S} \|\mu_\eta\| \leq \|\mu\|$ and $\sum_k |t|^{p_m(k)}$ is bounded uniformly in m on such subdiscs. The set $\{r_\alpha : \alpha \in A'\}$ has an accumulation point in $(b/2, b)$, hence $F \equiv 0$.

Compare coefficients. The sets $P_\infty, P_0, P_1, \dots$ are disjoint. On P_∞ , the same argument as in the one-block construction applies. Since the range of g has infinite complement, coefficients outside $\{0\} \cup \{g(\beta) + 1 : \beta \leq \gamma\}$ give $d = 0$. The coefficient at $p_\infty(0)$ gives $c = 0$, and the coefficients at $p_\infty(g(\beta) + 1)$ give $b_\beta = 0$ for all $\beta \leq \gamma$. Thus ν annihilates $\mathbf{1}$ and all u_ξ 's; these span a dense subalgebra of $C(B)$, so $\nu = 0$.

Fix $\eta \in S$. On the exponent set $P_{g(\eta)}$, only the η -gap contributes. Since $a_{g(\eta)} > 0$, the coefficients at $p_{g(\eta)}(g(\beta) + 1)$ give

$$b_{\eta,\beta} = c_\eta \quad (\beta \leq \gamma).$$

For $\beta = 0$, $b_{\eta,0} = 0$, because μ_η has no mass at the left endpoint of the model interval. Hence $c_\eta = 0$, and then all $b_{\eta,\beta} = 0$. Therefore $\mu_\eta(u_\xi) = 0$ for every $\xi < \omega_1$. Since μ_η has no endpoint masses, it annihilates the dense span of the endpoint-vanishing differences $u_\beta - u_\alpha$ in the quotient Z ; equivalently, the induced functional on Z is zero.

We have shown that μ annihilates $E[C(B)]$ and every gap summand in W . Since $C(L) = E[C(B)] \oplus W$, it follows that $\mu = 0$. Thus no nonzero functional vanishes on uncountably many x_α 's. By Lemma 2, $C([0, \omega_1^2])$ admits an overcomplete set. $\square$

5 Classification reduction below ω_2

We now combine the representative cases. We use the classical isomorphic classification of ordinal interval $C(K)$ -spaces due independently to Gul'ko–Os'kin and Kislyakov. In the form needed here, for ordinals of cardinality ω_1 :

- finite multiplicities $[0, \omega_1 n + \delta]$, $n < \omega$ and $\delta < \omega_1$, reduce to finite ℓ_∞ -sums of blocks of type at most $[0, \omega_1]$;
- all countably infinite multiplicities $\omega_1 \cdot \rho \leq \xi < \omega_1 \cdot (\rho + 1)$ with $\omega \leq \rho < \omega_1$ are isomorphic to the representative $C([0, \omega_1 \cdot \omega])$;
- all ξ with $\omega_1^2 \leq \xi < \omega_2$ are isomorphic to the representative $C([0, \omega_1^2])$.

This is the $\lambda = \omega_1$ instance of Kislyakov's classification; the same trichotomy is summarized in later ordinal-classification literature.

Theorem 4. *In ZFC, for every ordinal $\xi < \omega_2$, the Banach space $C([0, \xi])$ admits an overcomplete set.*

Proof. If $\xi < \omega_1$, then $C([0, \xi])$ is separable, so Klee's theorem gives an overcomplete sequence.

Let $\omega_1 \leq \xi < \omega_2$. Write

$$\xi = \omega_1 \cdot \rho + \delta, \quad \delta < \omega_1.$$

If $\rho < \omega$, the earlier finite-block argument applies: $[0, \xi]$ splits into finitely many clopen ordinal blocks of type at most $[0, \omega_1]$, and Koszmider's $C([0, \omega_1])$ criterion plus finite ℓ_∞ -sum stability give an overcomplete set.

If $\omega \leq \rho < \omega_1$, the classification quoted above gives

$$C([0, \xi]) \sim C([0, \omega_1 \cdot \omega]).$$

The countable-block packet proves that $C([0, \omega_1 \cdot \omega])$ admits an overcomplete set; equivalently, one may use the same sparse-exponent proof for the scalar plus countable c_0 -sum model. Since overcompleteness is preserved by Banach-space isomorphism, $C([0, \xi])$ admits an overcomplete set.

Finally, if $\omega_1 \leq \rho$, then $\xi \geq \omega_1^2$, and the same classification gives

$$C([0, \xi]) \sim C([0, \omega_1^2]).$$

By Theorem 3, $C([0, \omega_1^2])$ admits an overcomplete set. Therefore $C([0, \xi])$ admits an overcomplete set. $\square$

6 Verification and novelty check

Local indexes were searched for arXiv:2006.00806, Question 48, ordinal intervals, $C([0, \xi])$, $C([0, \omega_1 \cdot \omega])$, and $C([0, \omega_1^2])$. Existing lane-16 packets covered $\xi < \omega_1 \cdot \omega$ and then $C([0, \omega_1 \cdot \omega])$, but not the ω_1^2 -representative. Fresh literature searches for the exact question and close phrases did not locate a later paper explicitly resolving Koszmider's Question 48.

The proof relies on classical ordinal $C(K)$ -space classification and on the almost coherent injections already used in Koszmider's source paper. The new mathematical step is Theorem 3, especially the skeleton extension plus endpoint-vanishing gap perturbations.

7 Human review recommendation

Review as a candidate full solution. The main checks are:

1. continuity of the constructed x_α 's at limit skeleton points in $[0, \omega_1^2]$;
2. the decomposition $C([0, \omega_1^2]) = E[C(B)] \oplus W$;
3. the coefficient comparison on P_∞ and on the active gap sets $P_{g(\eta)}$;
4. the use of Kislyakov's classification to reduce all $\xi < \omega_2$ to the finite, countable, and ω_1^2 representatives.

References

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